

EXPERIMENT-21 Towers of Hanoi

Aim

To write a Prolog program to solve the **Towers of Hanoi** problem.

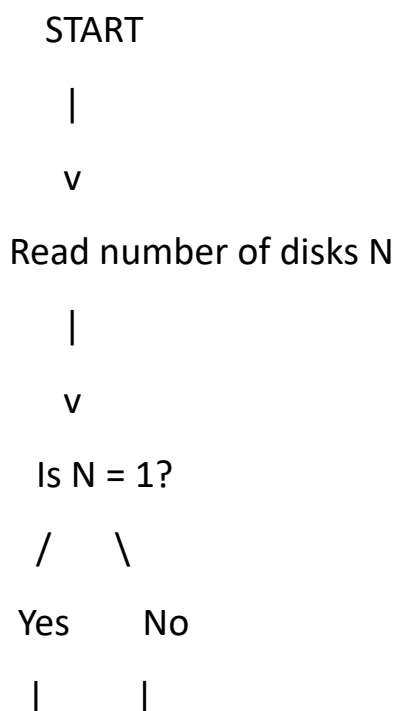
Objectives

- To understand recursion in Prolog.
- To implement the Towers of Hanoi problem using recursive rules.
- To generate the sequence of moves required to transfer disks.

Algorithm

1. If there is only one disk, move it directly from source to destination.
2. For $N > 1$:
 - Move $N-1$ disks from source to auxiliary.
 - Move the largest disk from source to destination.
 - Move $N-1$ disks from auxiliary to destination.
3. Repeat recursively until all disks are moved.

Flowchart



v v

Move disk Move N-1 disks

| Source -> Auxiliary

| |

| v

| Move largest disk

| |

| v

| Move N-1 disks

| Auxiliary -> Destination

| |

+-----+

|

v

STOP

Program

% Towers of Hanoi

hanoi(1, Source, Destination, _) :-

format('Move disk from ~w to ~w~n',
[Source, Destination]).

hanoi(N, Source, Destination, Auxiliary) :-

N > 1,

N1 is N - 1,

```

hanoi(N1, Source, Auxiliary, Destination),
format('Move disk from ~w to ~w~n',
      [Source, Destination]),
hanoi(N1, Auxiliary, Destination, Source).

```

Query

?- hanoi(3, left, right, center).

Output

File	View	AI Agent
<pre> HelloWorld.pl x :- initialization(main). % Towers of Hanoi hanoi(1, A, B, _) :- write('Move disk from '), write(A), write(' to '), write(B), nl. hanoi(N, A, B, C) :- N > 1, N1 is N - 1, hanoi(N1, A, C, B), write('Move disk from '), write(A), write(' to '), write(B), nl, hanoi(N1, C, B, A). main :- hanoi(3, left, right, middle). :- initialization(main). </pre>	<pre> STDIN Input for the program (optional) Output: Move disk from left to right Move disk from left to middle Move disk from right to middle Move disk from left to right Move disk from middle to left Move disk from middle to right Move disk from left to right Move disk from left to right Move disk from left to middle Move disk from right to middle Move disk from left to right Move disk from middle to left Move disk from middle to right Move disk from left to right </pre>	

Result

The Towers of Hanoi problem was successfully implemented using recursion in Prolog.

Conclusion

The program demonstrates how recursive rules can be used to solve a classical problem in Prolog.